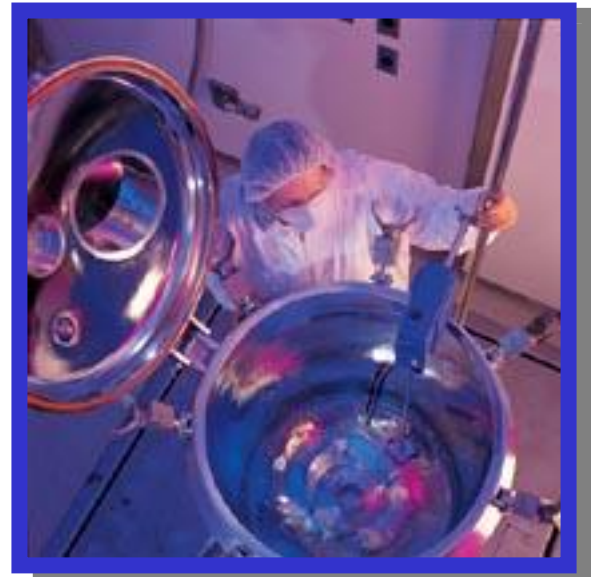
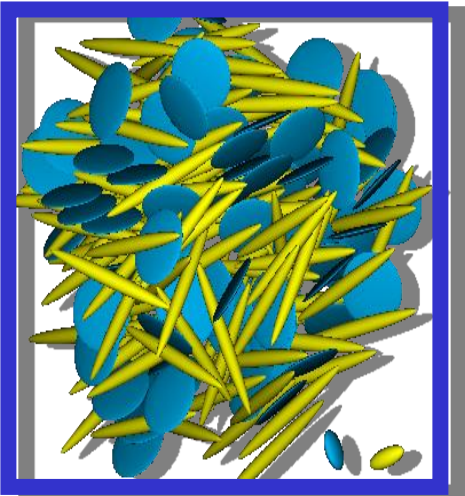
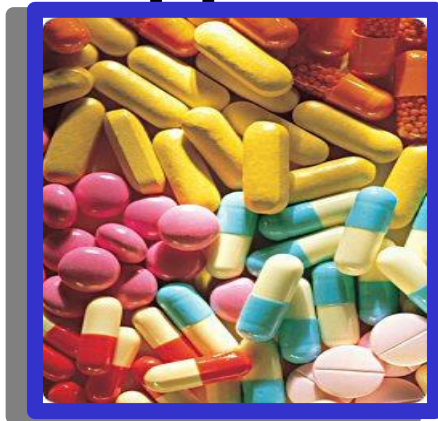


Protein Purification Techniques

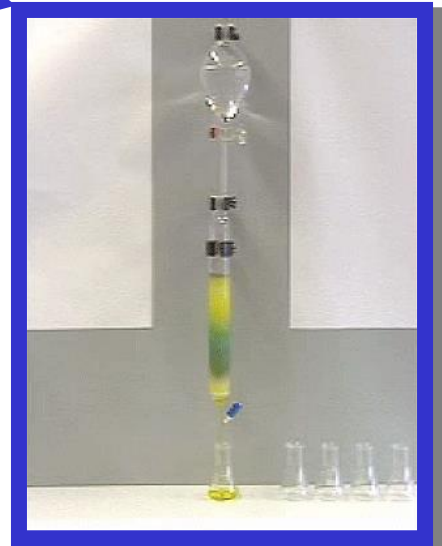
What is Chromatography?



Applications of Chromatography



Types of Chromatography...



- Before a protein can be analyzed, it must first be isolated in a pure state, i.e., **purified**
- Protein purification means separating the protein of interest from other proteins and components

1.Extraction

2.Precipitation

3.Purification

4.Concentration

5.Storage

Isolation of Proteins from Microbial Cells

- Proteins may be
 - in cell free supernatant
 - Ammonium sulphate precipitation
 - Membrane bound
 - NaCl extraction
 - Intracellular
 - Sonication

Ammonium sulphate: Salting out

- After Proteins solubilized, they can be purified based on solubility (usually dependent on overall charge, ionic strength, polarity)
- Ammonium sulfate (NH_4SO_4) commonly used to “salt out”
- Takes away water by interacting with it, makes protein less soluble because hydrophobic interactions among proteins increases
- Different aliquots taken as function of salt concentration to get closer to desired protein sample of interest (**30, 40, 50, 75% increments**)
- One fraction has protein of interest

- E.g. 30% Ammonium sulphate
- Mix the supernatant with salt
- Keep at 4°C overnight
- Cold centrifuge on next day
- Supernatant and pellet is obtained.

NaCl Extraction

- Suspend the culture in different concentrations of NaCl i.e. 1, 2, 3 and 4M for different time intervals (2 to 24h).
- These were incubated at 37°C under shaking conditions (150 rpm).
- The salt treated sample was centrifuged at 1500 rpm for 15 min.
- Cell debris was suspended in minimum amount of PBS. Both cell debris and supernatant were dialysed against PBS at 4°C overnight.

Sonication

- Sonication is a process in which sound waves are used to agitate particles in solution
- Place the tube on ice and immerse probe in the sample.
- Place the probe inside the ice for 20 seconds between samples. (avoid overheating)
- Till the sample gets clear.
- Centrifuge at 4°C

Dialysis

- Passage of solutes through a semi-permeable membrane.
- Pores in the dialysis membrane are of a certain size.
- Protein stays in; water, salts, protein fragments, and other molecules *smaller than the pore* size pass through.
- e.g. Cellophane (cellulose acetate)

(a) At start of dialysis

(b) At equilibrium

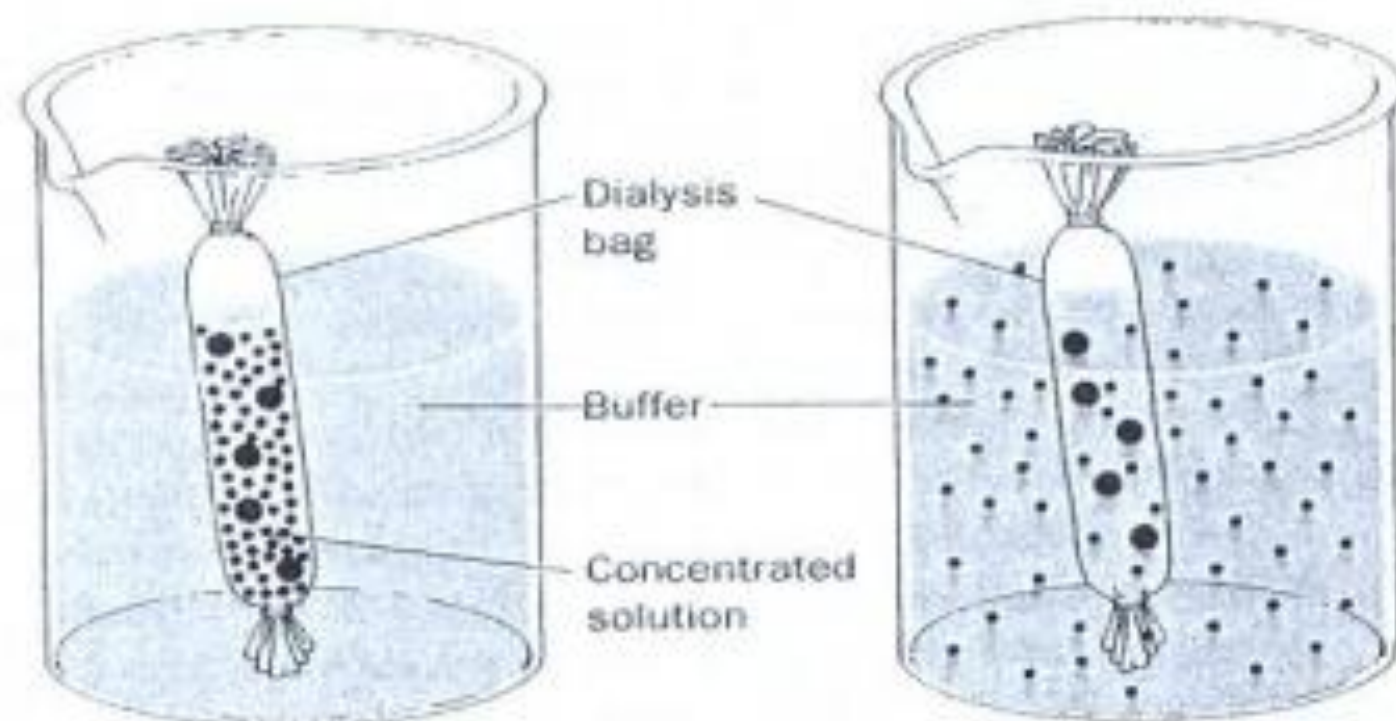


Figure 5-14

The separation of small and large molecules by dialysis. (a) Only small molecules can diffuse through the pores in the bag. (b) At equilibrium the concentrations of small molecules are nearly the same inside and outside the bag, whereas the macromolecules remain in the bag.

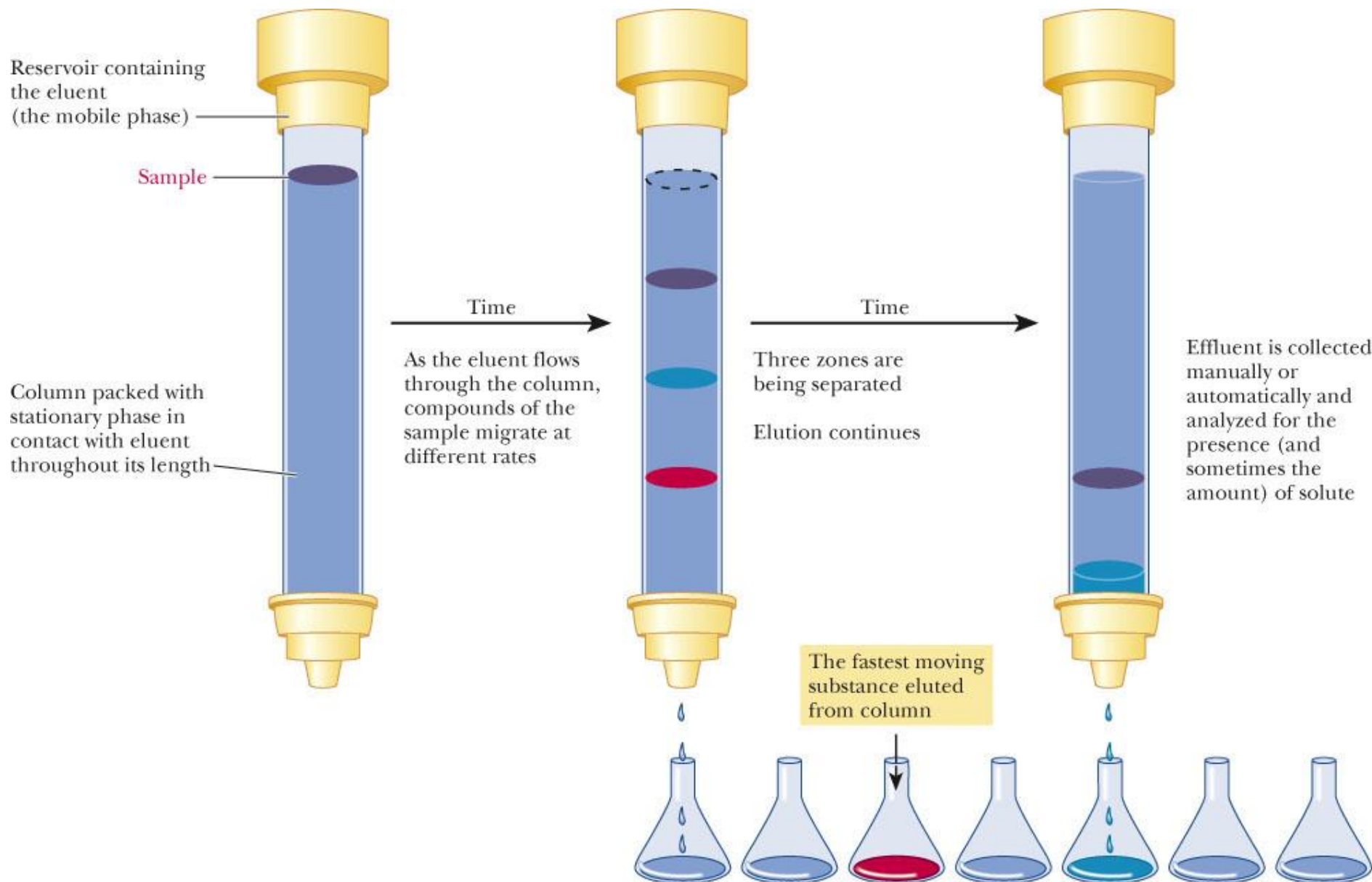
Concentration

- Sample in a dialysis bag can be coated with dry forms of the polymers, water will be removed from the dialysis bag and hydrate the polymer on the exterior.
- e.g. polyethylene glycols and polyvinyl pyrrolidones

Column Chromatography

- Basis of Chromatography
 - Different compounds distribute themselves to a varying extent between different phases
- Interact/distribute themselves
- In different phases
- **2 phases:**
 - Stationary: samples interacts with this phase
 - Mobile: Flows over the stationary phase and carries along with it the sample to be separated

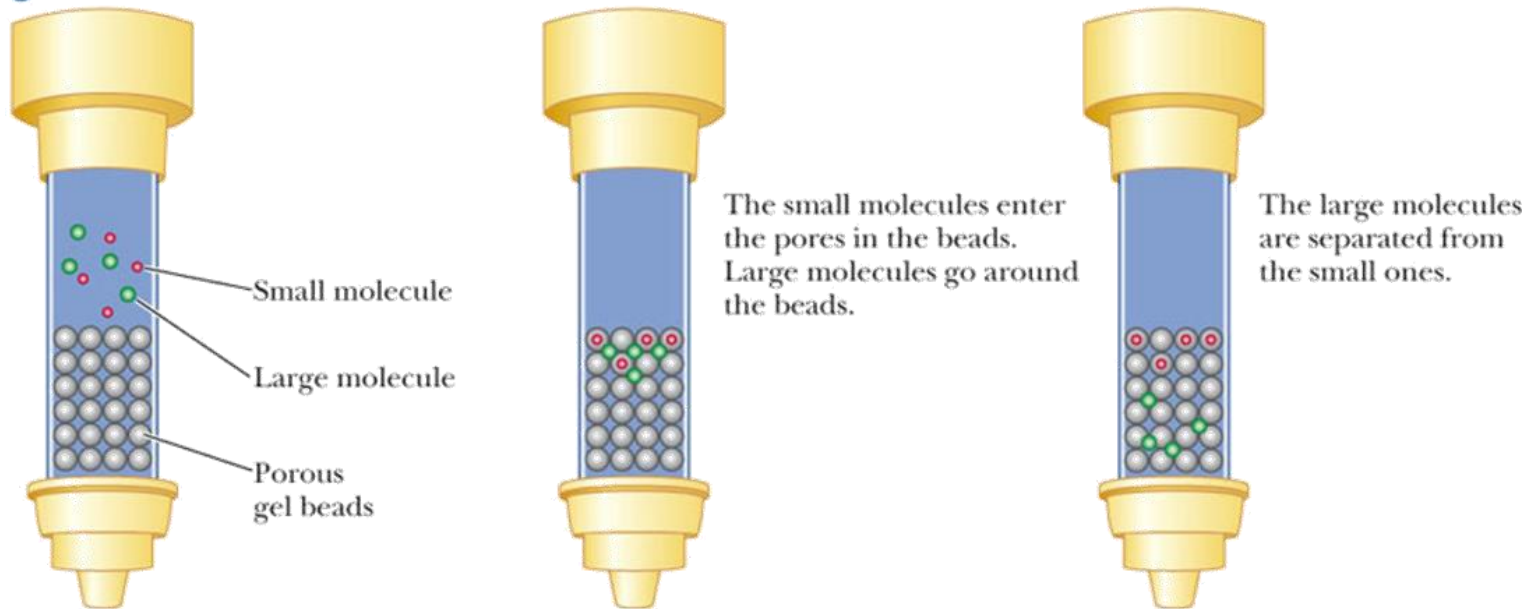
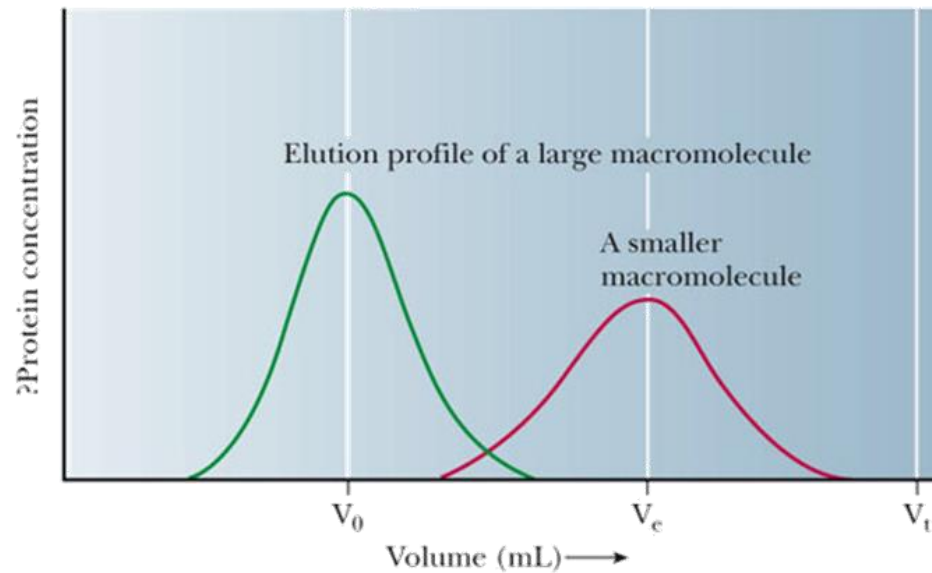
- Different kinds of chromatographic separations based on one of the following:
 - **size of protein** (molecular sieve chromatography = gel filtration = size exclusion chromatography), or
 - **net charge of protein** (ion exchange chromatography), or
 - **specific ligand binding** properties of protein (affinity chromatography)



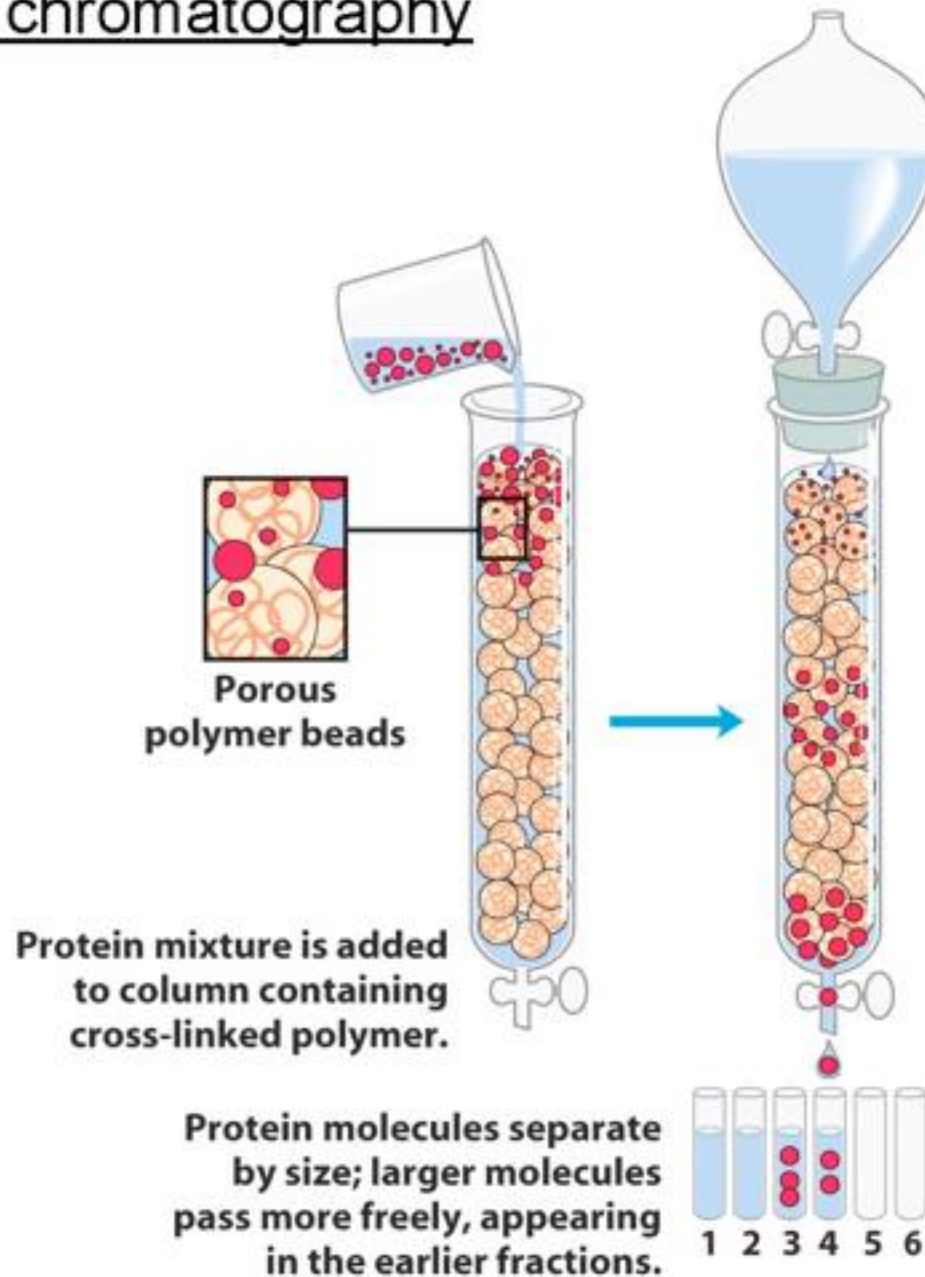
Size-Exclusion/Gel-Filtration/Molecular Sieve

- Separates molecules based on size.
- Stationary phase composed of cross-linked gel particles.
- Extent of cross-linking can be controlled to determine pore size
- Smaller molecules enter the pores and are delayed in elution time. Larger molecules do not enter and elute from column before smaller ones.

- Stationary phase (column matrix) = "beads" of a polysaccharide material that separates proteins based on **size** and **shape**.
- Column packing materials
 - hydrated, porous beads of carbohydrate polymer (e.g. dextran or agarose) or polyacrylamide)
 - wide range of molecular exclusion limits

A**B**

Size-exclusion chromatography

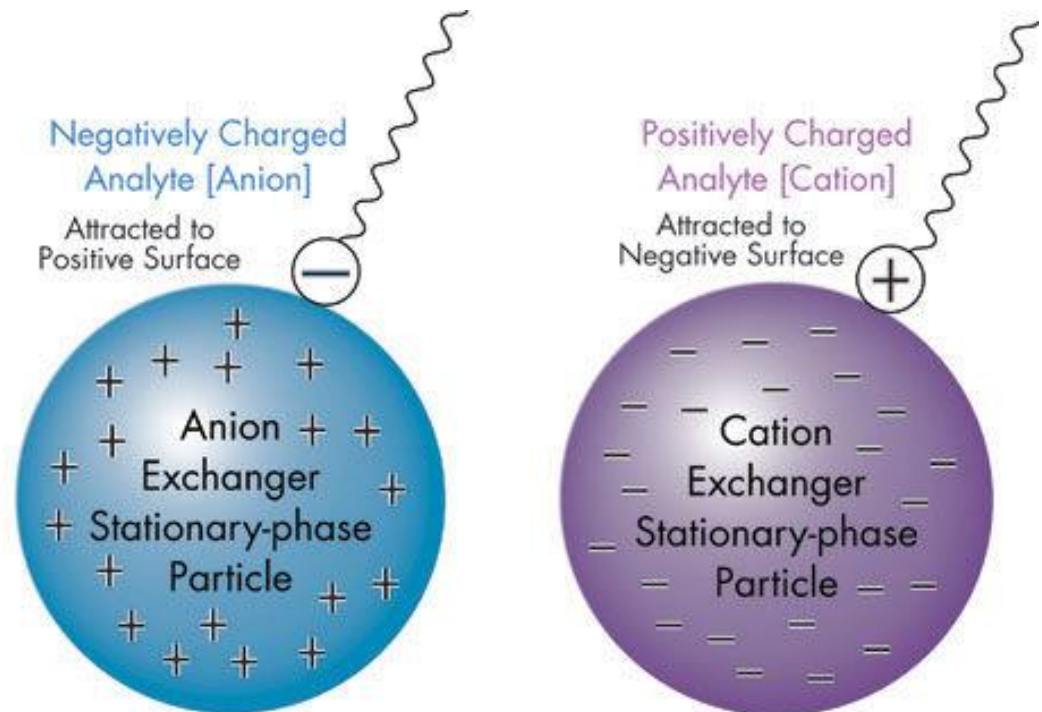


- **volume of buffer required to elute a specific protein depends mostly on molecular weight of the protein.**

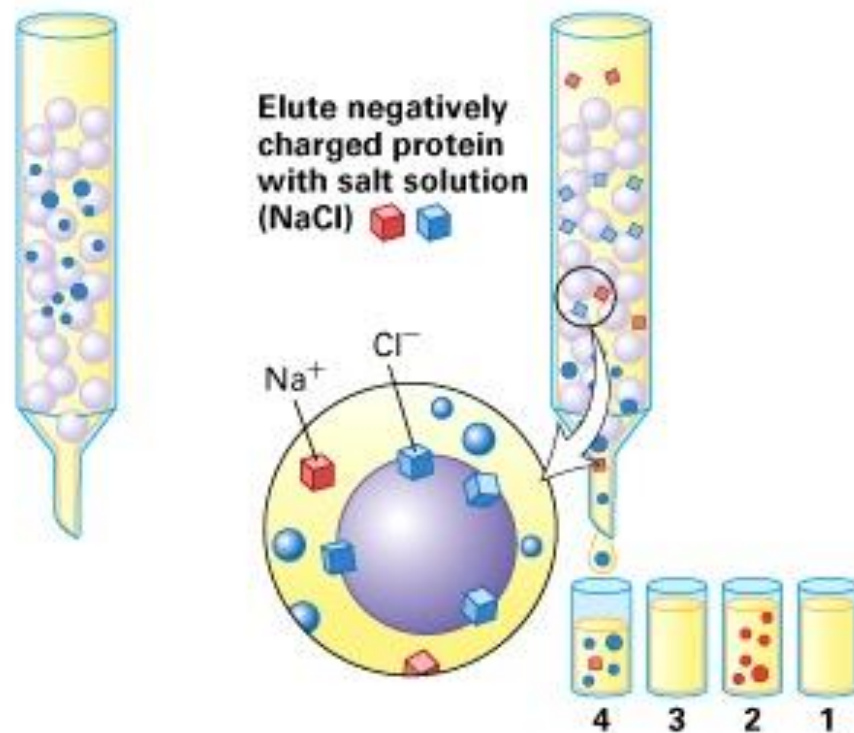
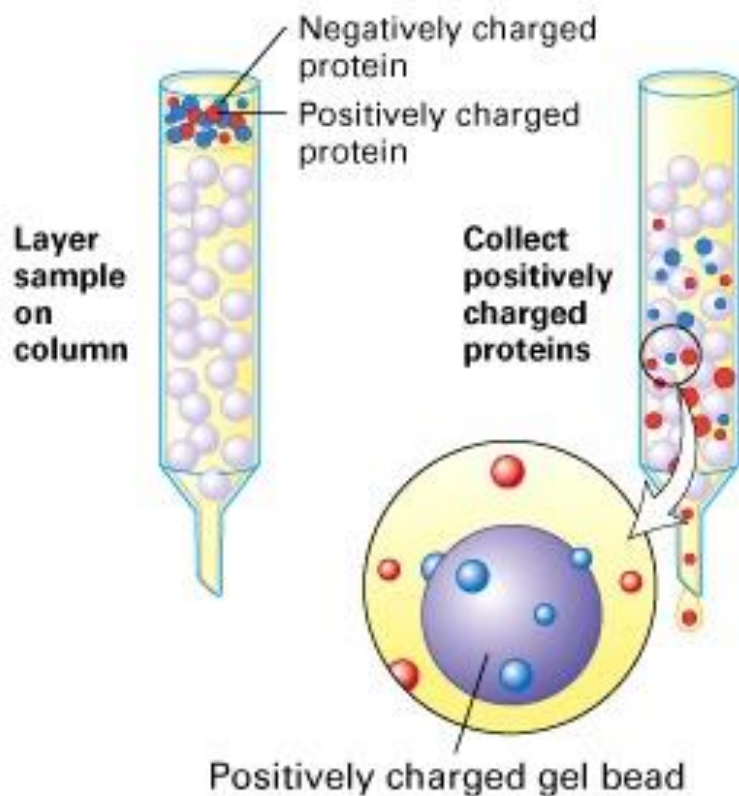
Ion Exchange Chromatography

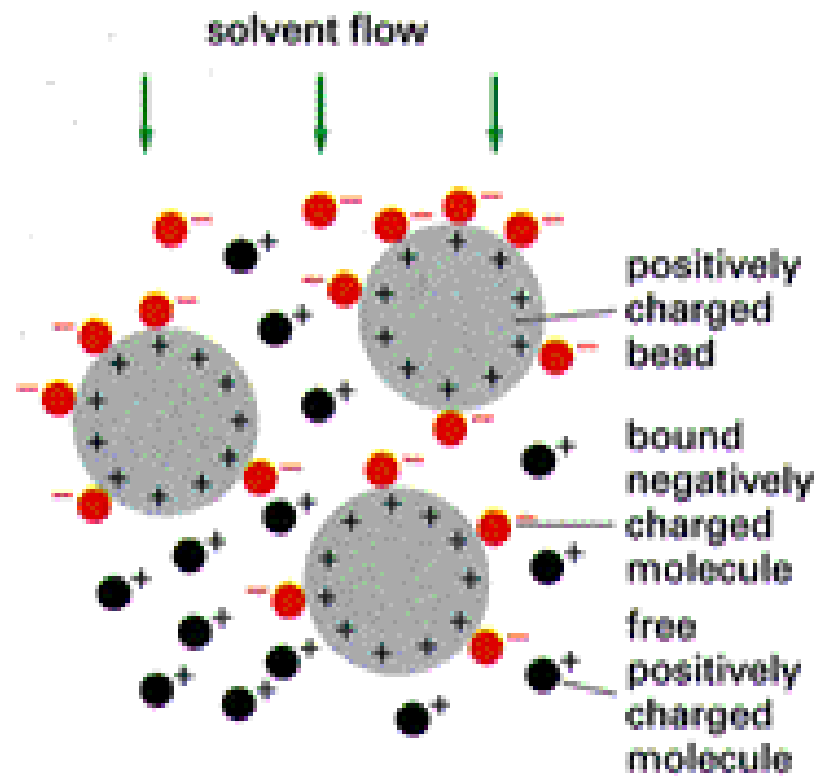
- resins have charged groups covalently attached to the stationary phase
 - positive or negative
- Proteins bind to the matrix by **electrostatic interactions**.
- Strength depends upon
 - Net charge on the protein
 - Salt concentration of the buffer

- Interaction based on overall charge (less specific than affinity)
- Cation exchange
e.g. carboxymethyl
- Anion exchange
e.g. DEAE



(b) Ion-exchange chromatography





(A) ION-EXCHANGE CHROMATOGRAPHY

Affinity Chromatography

- Uses specific binding properties of molecules/proteins
- Stationary phase has a **polymer** that can be covalently linked to a compound called a **ligand** that specifically binds to protein
- Matrix: cross-linked dextrans, polystyrene, polyacrylamide, cellulose, porous glass and silica.
- Ligands: antibodies, protein receptors, steroid hormones, vitamins, and certain enzyme inhibitors.

- **Spacer arm:** is added between the ligand and the matrix
 - reduces steric hindrance
 - 6-10 carbon atoms or their equivalent e.g. CNBr
 - neither chemically or structurally affect the sample or the ligand.
- **Ligand Coupling** : three steps
 - matrix is activated towards functional group of the ligand
 - ligand is covalently coupled through some chemical reaction
 - residual unreacted groups are blocked

- When a mixture of proteins is passed through the column, only those few that bind strongly to the ligand stick, while the others pass through the column
- Protein of interest is eluted with a buffer containing the free ligand, which competes with the column ligand to bind to the protein, and protein washes off
- Useful in: Igs, DNA binding proteins, receptor proteins, enzymes

Column with substance S
covalently bonded to
supporting material

Sample
containing
mixture of
proteins

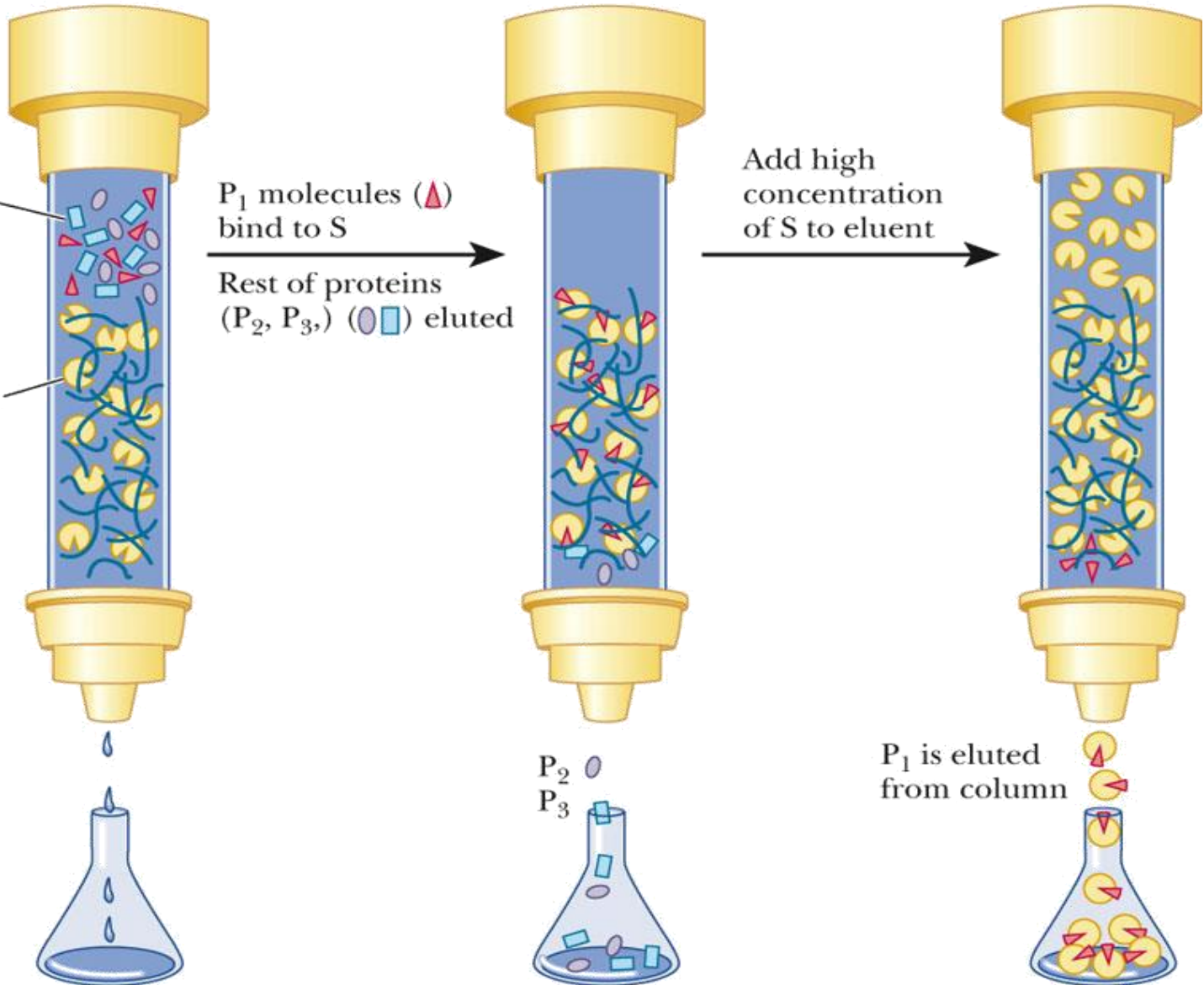
Substance S

P_1 molecules ($\blacktriangle$)
bind to S

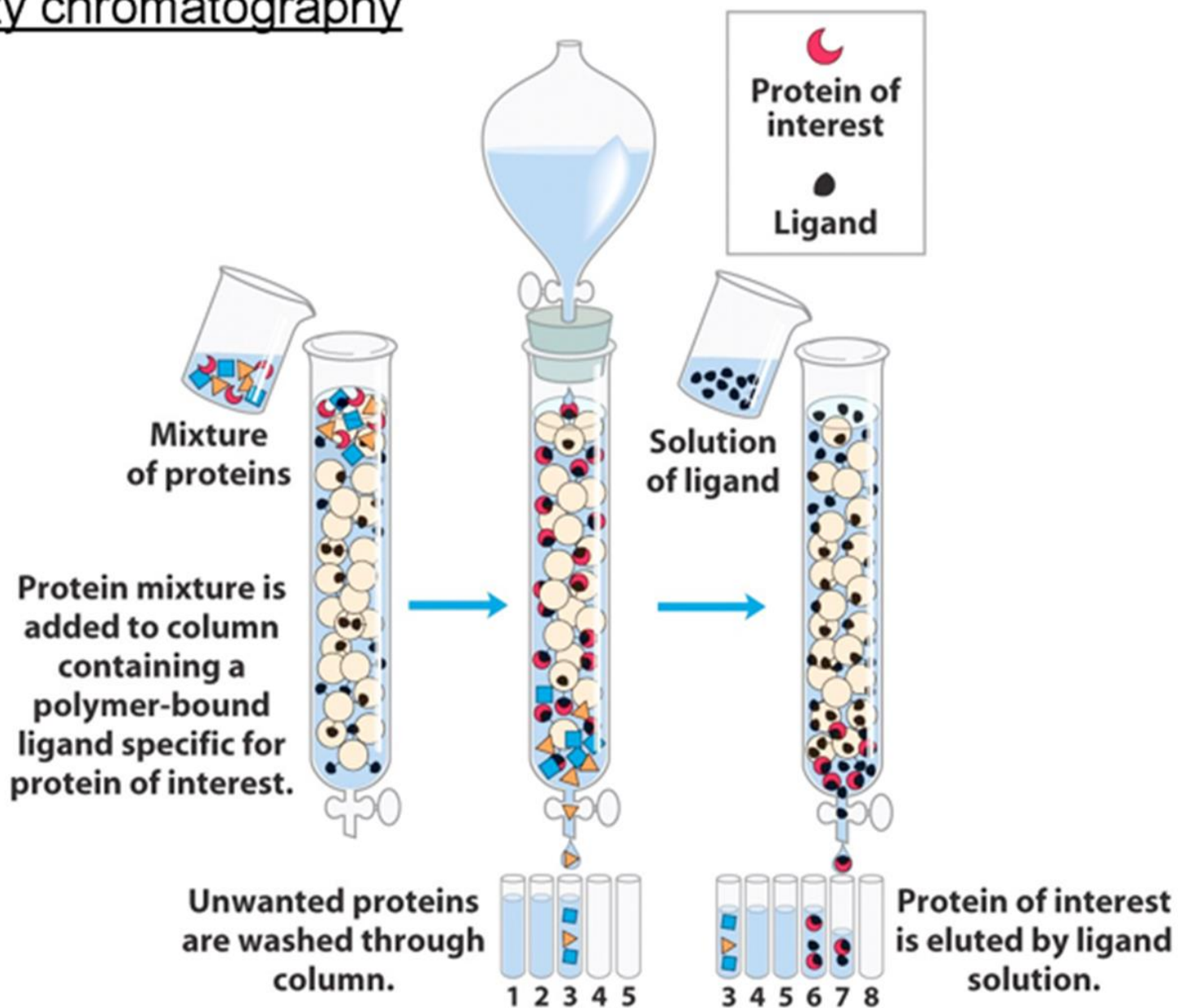
Rest of proteins
($P_2, P_3,$) ($\circ$ $\square$) eluted

Add high
concentration
of S to eluent

P_1 is eluted
from column

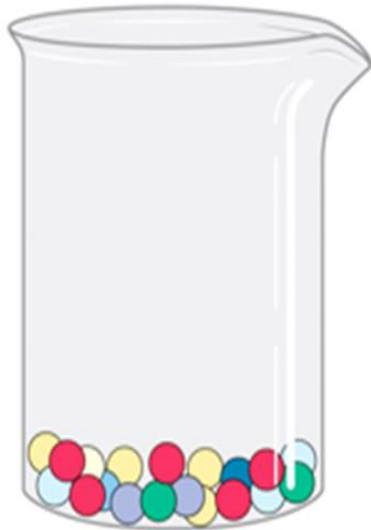
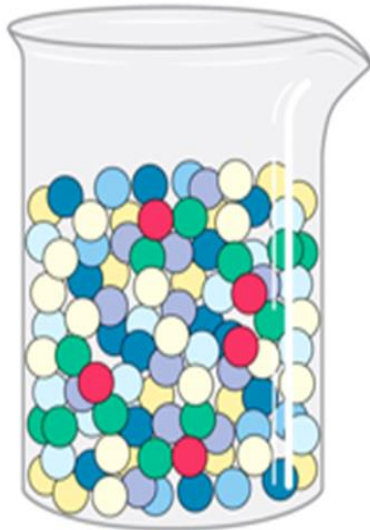


Affinity chromatography



Purification table of hypothetical of enzyme

<i>Procedure or step</i>	<i>Fraction volume (ml)</i>	<i>Total protein (mg)</i>	<i>Activity (units)</i>	<i>Specific activity (units/mg)</i>
1. Crude cellular extract	1,400	10,000	100,000	10
2. Precipitation with ammonium sulfate	280	3,000	96,000	32
3. Ion-exchange chromatography	90	400	80,000	200
4. Size-exclusion chromatography	80	100	60,000	600
5. Affinity chromatography	6	3	45,000	15,000

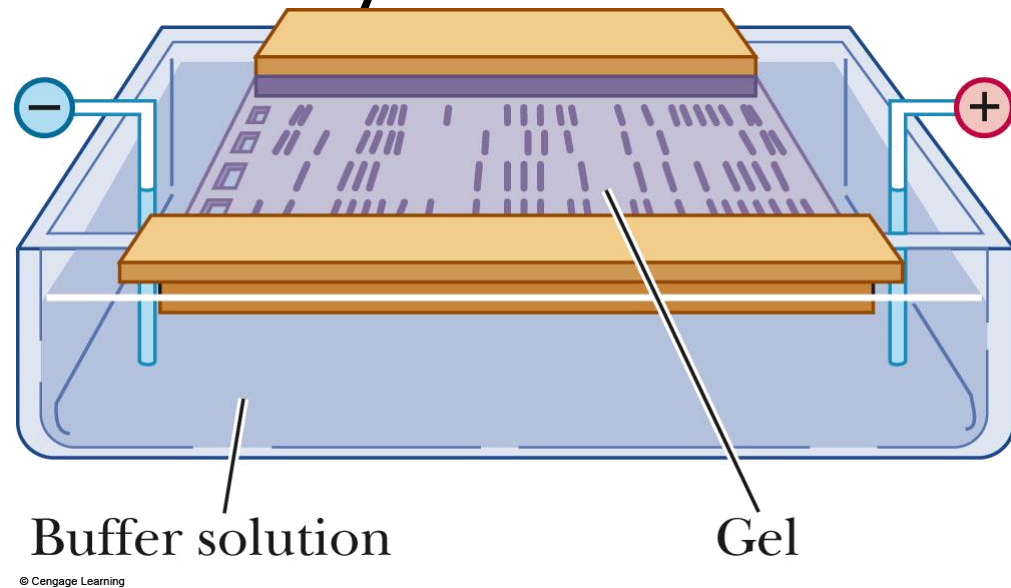


Specific activity=
enzyme activity / amount of protein
(mg)

Electrophoresis

- **Electrophoresis**- charged particles migrate in electric field toward opposite charge
- Proteins have different mobility:

- Charge
- Size
- Shape

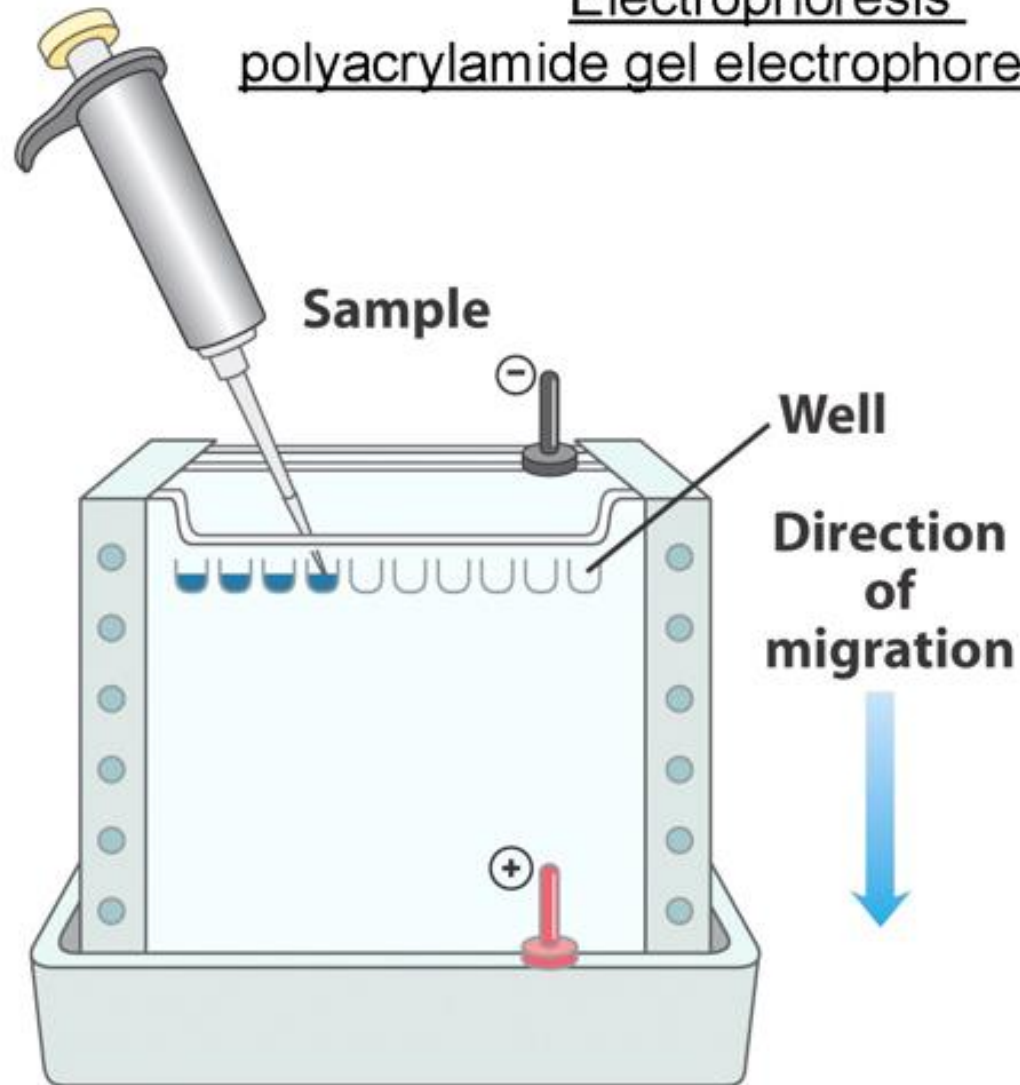


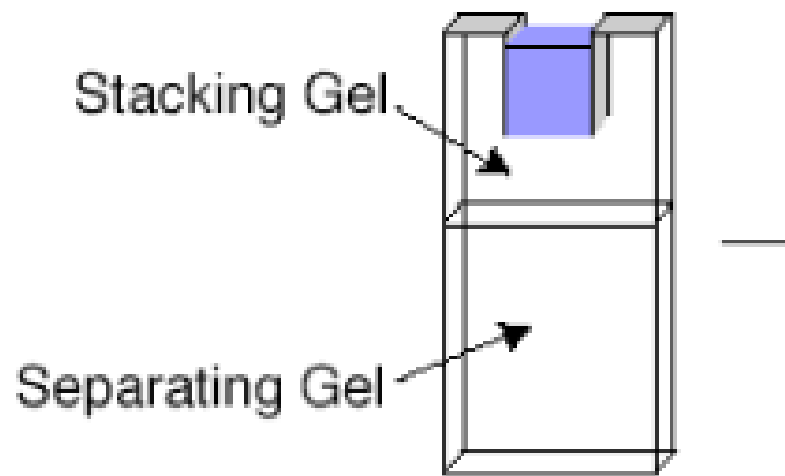
- Agarose used as matrix for nucleic acids
- Polyacrylamide used mostly for proteins

Electrophoresis (Cont'd)

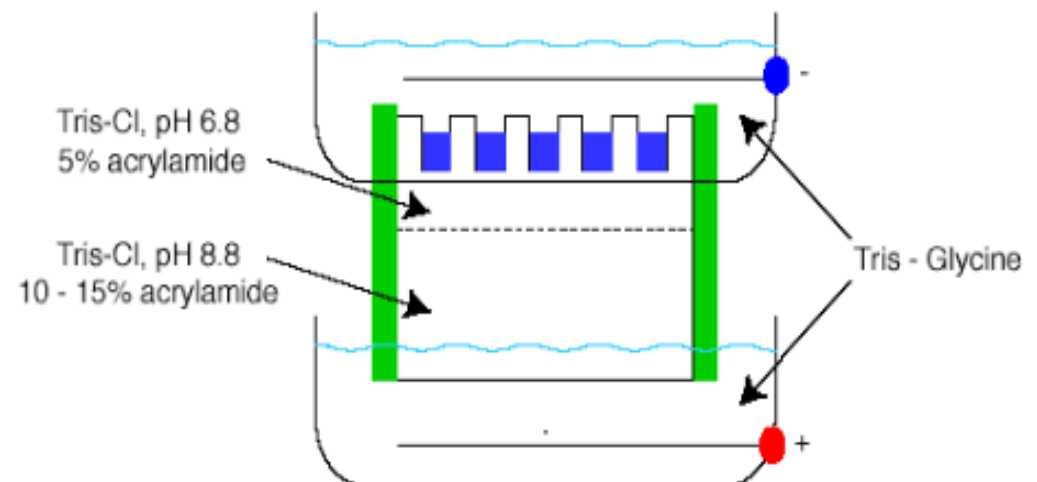
- Polyacrylamide has more resistance towards larger molecules than smaller
- Protein is treated with detergent (SDS) sodium dodecyl sulfate
- Smaller proteins move through faster (charge and shape usually similar)

Electrophoresis
polyacrylamide gel electrophoresis (PAGE)





- **Buffer compositions control stacking and separation:**

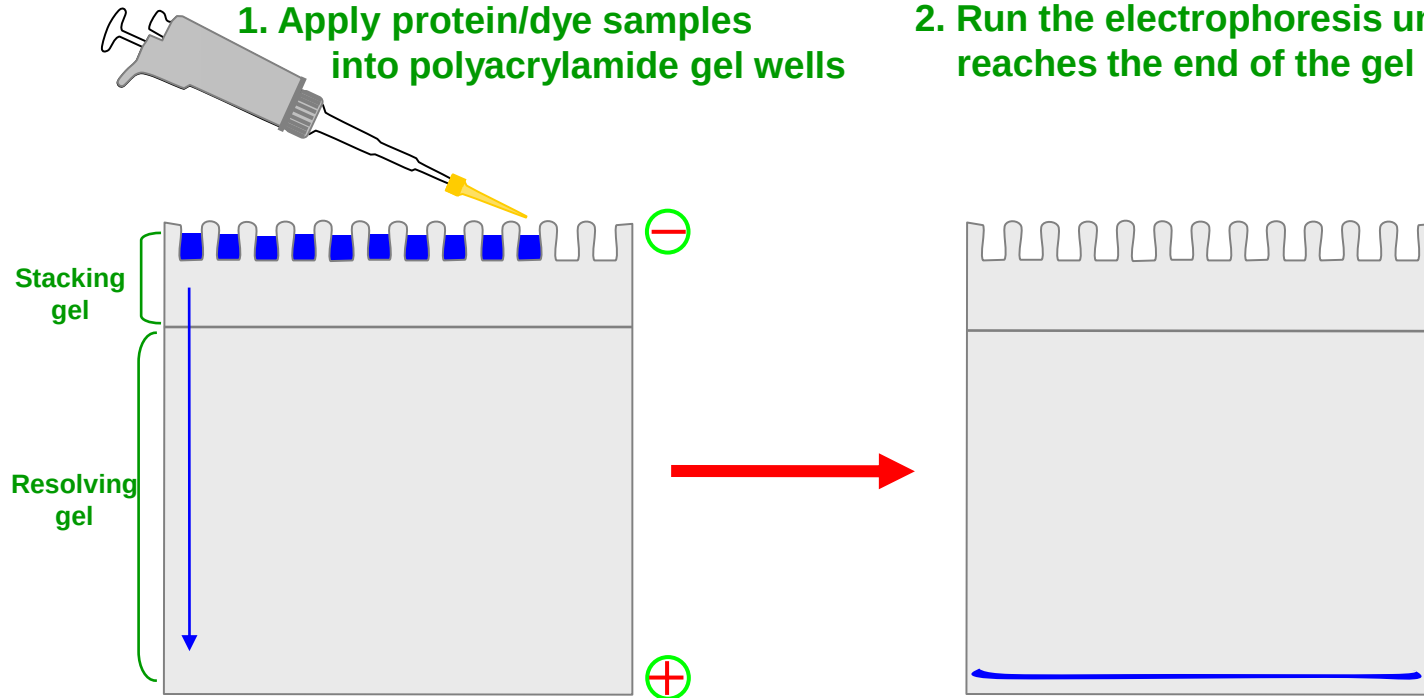


Protein separation using SDS-PAGE

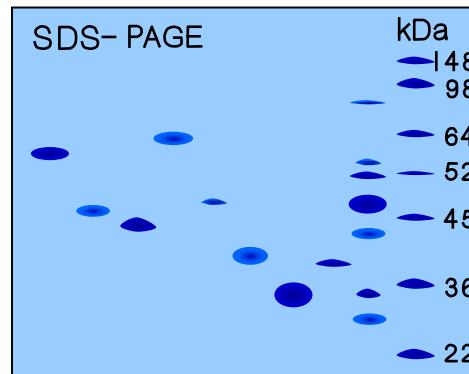
(Laemmli system)

1. Apply protein/dye samples into polyacrylamide gel wells

2. Run the electrophoresis until dye reaches the end of the gel



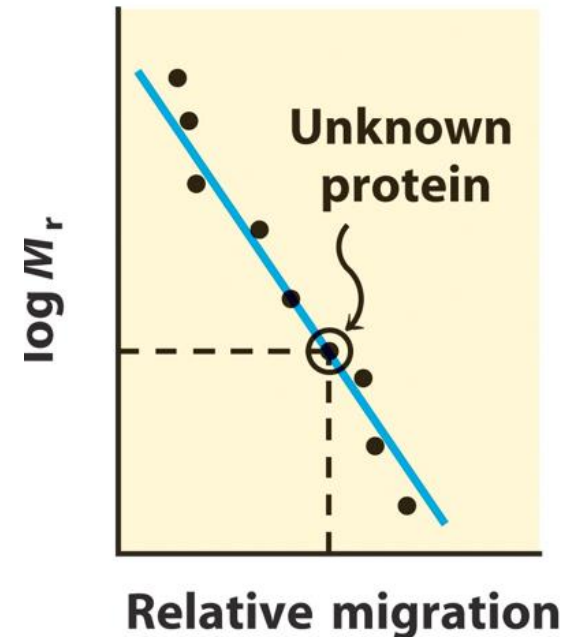
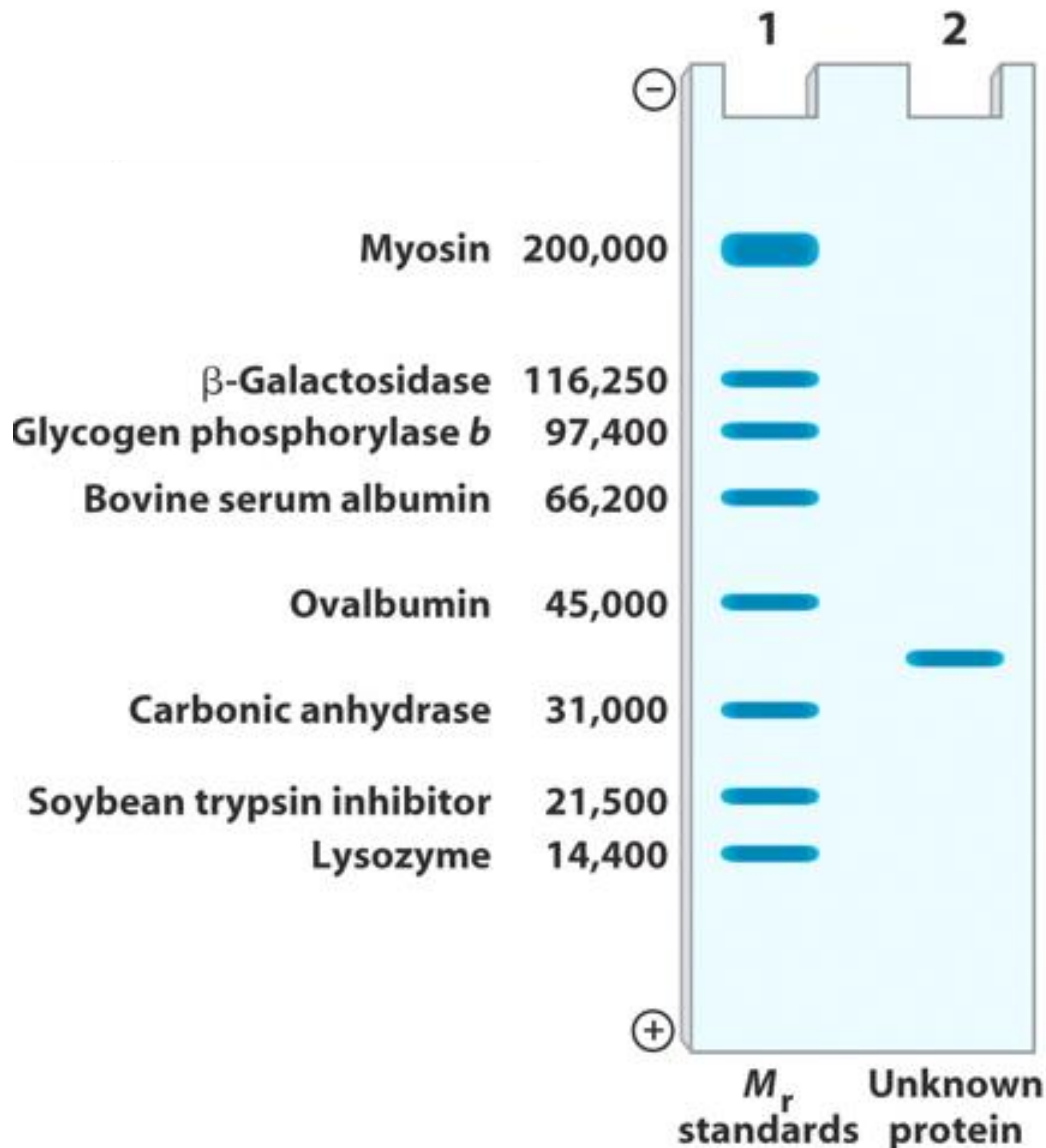
3. Remove the gel from the apparatus and stain for proteins



SDS-PAGE

- Most proteins bind SDS at a constant ratio of about 1.4 g SDS/g protein, i.e., about 1 SDS for every 2 amino acid residues, unfolding the proteins.
- Sample treatment before running gel included beta-mercaptoethanol reduction (so no disulfide bonds left) and heating to ensure complete unfolding and complete separation of different polypeptide chains

Determining Mass of Polypeptide



Storage

- For short term storage (**up to 24 h**), most proteins can be kept at 4°C.
- For long term storage (**more than a week**), it becomes necessary to freeze the protein preparation
- Protein preparations will be stable for several **years** at -80°C or even in liquid nitrogen
- It is also important to freeze the solution in small aliquots to avoid repeated freezing and thawing which may reduce the biological activity or affect the structure.